

Web Information Retrieval

Exercise Sheet

February 12, 2022

Exercise 1 – The Anatomy of a Web Search Engine

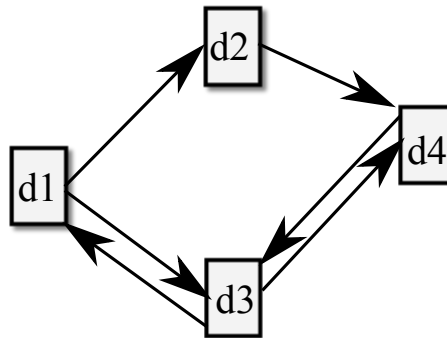
Explain in your own words how a Web search engine basically works, by discussing its architectural components as presented in the lecture. Explain what each component is used for and how they relate to each other.

Exercise 2 – PageRank and the Random Surfer Model

Describe how the iterative PageRank formula introduced in the lecture implements the Random Surfer Model!

Exercise 3 – Link Analysis 1

Let the following hypertext graph be given:



- a) Compute the PageRank of all documents (with $\beta = 0.8$). Finish after 2 iterations. The initial PageRank of each node is 1.
- b) Compute hub and authority values for each document using the HITS algorithm. Finish after 2 iterations. The initial hub and authority value of each node is 1.

Exercise 4 – Link Analysis 2

Discuss the link analysis approaches we presented in the lecture in your own words. What are their strengths and weaknesses? How can their output possibly be combined with the content-based *RSV* computed with the vector space model?